



Institute of Computer Engineering Technology

iCD iCET Certified Developer

ASSIGNMENT

Module	DataBase Management System
Assignment Title	DML
Assignment Type	Practical Exercise

Activity 01

Database Management for Iron & Oil Custom Garage

Business Scenario:

You are the lead software engineer developing the backend database for "Iron & Oil Custom Garage", a popular workshop that modifies and repairs motorcycles. The garage uses three database tables to manage its daily operations: Riders (Customers), Mod_Parts (Inventory), and Workshop_Jobs (Service records).

Your task is to execute a series of database operations to update the system at the end of the day.

Current Database Schema and Data Overview

The database comprises three key tables detailing information about riders, modification parts, and workshop jobs.

1. Riders Table

This table stores details for each member, including their contact information and membership status.

Rider ID	Full Name	Contact Number	Membership Tier
1	Alex Fernando	071-1111111	Standard
2	Sarah Silva	077-2222222	Standard
3	Chris Perera	070-3333333	VIP

2. Mod_Parts Table

This table catalogs the available modification parts, including their pricing and current inventory levels.

Part ID	Part Name	Category	Price (LKR)	Stock Quantity
201	LED Aux Lights	Electrical	4,500.00	15

202	Crash Bars	Protection	8,500.00	4
203	Sport Exhaust	Performance	12,000.00	2
204	Custom Grips	Accessories	1,500.00	25

3. Workshop_Jobs Table

This table tracks ongoing and completed workshop tasks, linking them to a specific rider and the part being installed, along with the labor cost and job status.

Job ID	Rider ID	Part ID	Labor Charge (LKR)	Job Status
5001	1	202	2,000.00	In Progress
5002	2	201	1,500.00	Pending
5003	3	203	3,500.00	Completed

Answer the following DML questions,

1. A new customer just arrived at the garage. Insert a new record into the Riders table: rider_id = 4, full_name = 'Daniel Raj', contact_no = '076-4444444', and membership_tier = 'Standard'.
2. A new shipment of parts has arrived. Insert a new item into the Mod_Parts table: part_id = 205, part_name = 'Touring Backrest', category = 'Comfort', price = 6500.00, and stock_quantity = 10.
3. Daniel (rider 4) wants to install the new Touring Backrest (part 205). Create a new record in the Workshop_Jobs table: job_id = 5004, rider_id = 4, part_id = 205, labor_charge = 1000.00, and job_status = 'Pending'.
4. The mechanic just finished installing Sarah's LED lights. Update the Workshop_Jobs table to change the job_status of job_id 5002 to 'Completed'.

5. The garage is running a promotion upgrading specific loyal customers. Update the Riders table to change the membership_tier to 'Pro-Rider' for customers with rider_id 1 and 2. *(Must use the IN operator)*
6. There is a massive clearance sale on items with high stock. Update the Mod_Parts table to reduce the price to 1000.00 for ALL parts where the stock_quantity is strictly greater than (>) 20.
7. A job was entered by mistake and needs to be completely removed. Delete the record from the Workshop_Jobs table where the job_id is exactly 5001.
8. The supplier has stopped making specific performance parts. Delete the parts from the Mod_Parts table where the part_id is 203 or 204. *(Must use the IN operator)*.
9. As part of an annual system cleanup, the manager wants to waive off and remove tiny jobs. Delete ALL records from the Workshop_Jobs table where the labor_charge is strictly less than (<) 1200.00.

Activity 02

IT Support Tickets - Azure Heights Hotel

Instructions for Students:

You are the Database Administrator for the "Azure Heights Hotel". Instead of direct database commands, you receive daily requests from different hotel departments via the internal ticketing system. Read the messages below and write the correct SQL queries to fulfill their requests.

You have access to the following three tables:

1. **Guest_Profiles** * Columns: guest_id (Number), full_name (Text), email (Text), reward_points (Number)
2. **Room_Inventory** * Columns: room_no (Number), category (Text), nightly_rate (Number), cleaning_status (Text)
3. **Bookings** * Columns: booking_id (Number), guest_id (Number), room_no (Number), booking_status (Text)

IT Helpdesk Inbox (Your Tasks)

Ticket #1001 - From: Front Desk Manager

“Hi IT Team, we have a walk-in VIP guest. Please create a new profile for him in the system. His name is 'Michael Scott', email is 'm.scott@paper.com', and give him a starting bonus of 500 reward points. Please assign him the guest_id 805. After that, book him into room 402 with booking_id 9901 and set the booking_status to 'Checked-In'. ”

Ticket #1002 - From: Head of Housekeeping

“Hello! My team just finished deep cleaning the entire top floor suites. Can you please update the system so that the cleaning_status for rooms 501, 502, and 505 is now set to 'Ready'? Thanks!”

Ticket #1003 - From: Sales Director

“Urgent request! We are running a weekend flash sale. Please go into the Room Inventory and lower the nightly_rate to 85.00 for ALL rooms that belong to the 'Standard' category. Needs to be done ASAP”

Ticket #1004 - From: IT Security System (Automated Alert)

“[WARNING]: Fraudulent activity detected. Fake reservations were made using stolen credit cards. Immediate action required: Delete the bookings with booking_id 9044 and 9045 entirely from the database.”

Ticket #1005 - From: Loyalty Program Manager

“Hi DB Admin, we are doing a database cleanup. Due to a system bug last year, some profiles have negative reward points. Please permanently delete ALL guest profiles from the system where their reward_points are strictly less than (<) 0. Thank you.”

Submission Guidelines

1. Database Structure Screenshot

When submitting the assignment, include a screenshot of the database structure showing all the tables you created in the database. This screenshot should clearly display the table names and their columns.

2. Submission Format

The assignment must be submitted in **Document format**:

- PDF document (.PDF)

